

Ho Fung Tsoi

Postdoctoral Researcher

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Last updated: Sep 2026

Research Interests

Building machine learning methods for accelerating scientific discoveries at particle physics experiments.

Experimental particle physics: new physics searches (SUSY, exotic Higgs) and trigger system at the CERN LHC

Machine learning: self-supervised learning, foundation models, anomaly detection, symbolic regression

Fast inference on FPGAs: low-latency (sub-microsecond) machine learning algorithms on FPGAs

Education

University of Wisconsin–Madison, Ph.D. Physics 2024

Thesis: [Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC](#) (CMS Thesis Award)

Advisor: Sridhara Dasu

The Chinese University of Hong Kong, B.Sc. Physics 2018

Thesis: *First Passage Time Problem of the Time-Dependent Ornstein-Uhlenbeck Process*

Advisor: Chi-Fai Lo

Visiting student at UC Berkeley & Berkeley Lab (Jan – Aug 2017)

Positions

Postdoctoral Researcher, University of Pennsylvania (Supervisor: Dylan Rankin) 2024 – Present

Awards

CERN, [CMS PhD Thesis Award](#) 2024

UWisconsin, Cornelius P. and Cynthia C. Browne Fellowship 2024

CUHK, CN Yang Scholarship 2017

CUHK, Wei Lun Foundation Exchange Scholarship 2017

CUHK, Professor Dennis Yam Kuen Lo Physics Award 2016

Selected Research Experience

Search for new physics at the LHC

- **SUSY.** ATLAS Run 3 search for electroweak production of supersymmetric particles in compressed mass spectra with low-momentum tracks (2026–Present).
- **Exotic Higgs.** CMS Run 2 search for exotic Higgs boson decays into pseudoscalars in the final state of two b quarks and two τ leptons, and the combination with the $\mu\mu b\bar{b}$ final state [4] (2020–2024).

Novel ML methods for high energy physics

- **Self-supervised learning.** Self-distillation pre-training method **jBOT** for jets [14]; VICReg pre-training to improve resonance mass width reconstruction [15]; data-driven formation of invariant training pairs using energy mover's distance [16] (2025–Present).
- **SymbolFit.** Automatic discovery of parametric functions to model HEP data [6,13] (2024–2026).

Low-latency (sub-microsecond) ML algorithms on FPGAs

- **hls4ml.** Co-developer; an open-source tool that converts and optimizes trained neural networks into FPGA firmware [7] (2023–Present).
- **SpikePID.** Neuromorphic spiking neural networks (SNNs) for on-detector dN/dx clustering counting in future drift chambers [11] (2025–Present).
- **SparsePixels.** CNNs with sparse convolution for spatially sparse image data on FPGAs [8] (2024–Present).
- **SymbolNet.** Compression of neural networks using symbolic regression for fast inference [3,5] (2022–2024).

Trigger system at the LHC

- **GNTau.** Commissioning the GNN-based tau trigger at the ATLAS High-Level Trigger for Run 3 [9] (2024–2025).
- **CICADA.** Development of an anomaly detection trigger algorithm at the CMS Level-1 Trigger [10] (2022–2024).
- Development of the Data Quality Monitoring (DQM) software for the Calorimeter Layer-1 subsystem at the CMS Level-1 Trigger (2020–2024).
- HL-LHC trigger sensitivity projection for the CMS searches for exotic Higgs boson decays [1] (2020–2021).

Selected Publications

Listed below are selected publications to which I was a primary author. I also co-authored papers by [CMS](#) and [ATLAS](#) as a collaboration member. Complete list at [INSPIRE](#). Preprints at [arXiv](#).

2026

16. [Similarity Pairing with Energy Mover's Distance for Self-Supervised Pre-Training at the LHC](#)
[Ho Fung Tsoi](#), Dylan Rankin Submitted for review
[arXiv:2609.17738](#)
15. [Self-Supervised Learning for Robust Resonance Mass Regression in Cascade Decays](#)
[Ho Fung Tsoi](#), Alex Yang, Luis Felipe Gutierrez Zagazeta, Shion Chen, Dylan Rankin Submitted for review
[arXiv:2609.17726](#)
14. [jBOT: Semantic Jet Representation Clustering Emerges from Self-Distillation](#)
[Ho Fung Tsoi](#), Dylan Rankin SciPost Phys.
[arXiv:2601.11719](#)
13. [Machine Can Automatically Discover Parametric Functions to Model HEP Data](#)
[Ho Fung Tsoi](#), Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles ICHEP 2026
[arXiv:2607.19750](#)
12. [SparsePixels++: Scalable Sparse Convolution on FPGAs](#)
[Ho Fung Tsoi](#), Dylan Rankin, Philip Harris Preparing for sub.
11. [SpikePID: Real-Time Neuromorphic Cluster Counting for \$dN/dx\$ in Future Drift Chambers](#)
[Ho Fung Tsoi](#), Kam Wai Lai, Dylan Rankin Preparing for sub.
10. [Unsupervised Anomaly Detection for Real-Time Data Acquisition in the CMS Experiment at the LHC](#)
CMS Collaboration Preparing for sub.
CMS PAS
9. [Performance of the ATLAS Tau Trigger in Run 3](#)
ATLAS Collaboration Preparing for sub.

2025

8. [SparsePixels: Efficient Convolution for Sparse Data on FPGAs](#)
[Ho Fung Tsoi](#), Dylan Rankin, Vladimir Loncar, Philip Harris Mach. Learn. Sci. Tech.
[arXiv:2512.06208](#)
7. [hls4ml: A Flexible, Open-Source Platform for Deep Learning Acceleration on Reconfigurable Hardware](#)
Jan-Frederik Schulte et al. ACM TRETS
[arXiv:2512.01463](#)

2024

6. [SymbolFit: Automatic Parametric Modeling with Symbolic Regression](#)
[Ho Fung Tsoi](#), Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles, Vladimir Loncar Comput. Softw. Big Sci.
[arXiv:2411.09851](#)
5. [SymbolNet: Neural Symbolic Regression with Adaptive Dynamic Pruning for Compression](#)
[Ho Fung Tsoi](#), Vladimir Loncar, Sridhara Dasu, Philip Harris Mach. Learn. Sci. Tech.
[arXiv:2401.09949](#)
4. [Search for Exotic Decays of the Higgs Boson to a Pair of Pseudoscalars in the \$\mu\mu b\bar{b}\$ and \$\tau\tau b\bar{b}\$ Final States](#)
CMS Collaboration Eur. Phys. J. C
[arXiv:2402.13358](#)

2023

3. [Symbolic Regression on FPGAs for Fast Machine Learning Inference](#)
[Ho Fung Tsoi](#), Adrian Alan Pol, Vladimir Loncar, Ekaterina Govorkova, Miles Cranmer, Sridhara Dasu, Peter Elmer, Philip Harris, Isobel Ojalvo, Maurizio Pierini CHEP 2023
[arXiv:2305.04099](#)
2. [Searches for exotic Higgs boson decays with the CMS experiment](#)
[Ho Fung Tsoi](#) on behalf of the CMS Collaboration EPS-HEP 2023

2021

1. [The Phase-2 Upgrade of the CMS Data Acquisition and High Level Trigger](#)
CMS Collaboration CMS Technical Design Report

Conference, Workshop, Seminar Presentations

Fast Machine Learning for Science Conference (FastML) — UCSD, USA SpikePID: Neuromorphic Cluster Counting in Future Drift Chambers SparsePixels++: Scalable Sparse Convolution on FPGAs	Sep 2026
International Conference on High Energy Physics (ICHEP) — Natal, Brazil Machine Can Automatically Discover Parametric Functions to Model HEP Data	Jul 2026
TREASURE: Tokenizing HEP Collider Data for AI Discovery — BNL, USA Self-Supervised Learning: Machine Can Learn a Lot by Just Observing	Apr 2026
Coordinating Panel on Advanced Detectors (CPAD) — UPenn, USA SparsePixels: Efficient Convolution for Sparse Data on FPGAs	Oct 2025
2024 CMS Thesis Award Ceremony Talk (CMS Week) — CERN, Switzerland Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC	Oct 2025
Fast Machine Learning for Science Conference (FastML) — ETH Zurich, Switzerland SparsePixels: Efficient Convolution for Sparse Data on FPGAs	Sep 2025
High-energy physics seminar — UPenn; U.Washington; Fermilab, USA Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC	2023, 2024
US LHC Users Association Meeting (US LUA) — Fermilab, USA CICADA: Anomaly Detection for New Physics Searches at the CMS Level-1 Trigger	Dec 2023
Machine Learning at Level-1 Trigger Workshop (ML@L1) — CERN, Switzerland Anomaly Detection – CICADA: Status, Plans, and Prospects for Phase-2	Dec 2023
CMS Machine Learning Town Hall — CERN, Switzerland L1 Anomaly Detection with Calorimeter Inputs: Status and Opportunities	Sep 2023
European Physical Society Conference on High Energy Physics (EPS-HEP) — Hamburg, Germany Searches for exotic Higgs boson decays with the CMS experiment	Aug 2023
International Conference on Computing in High Energy & Nuclear Physics (CHEP) — Norfolk VA, USA Symbolic Regression on FPGAs for Fast Machine Learning Inference	May 2023
International Conference on Applied Mathematics — CityU, Hong Kong First Passage Time Problem of the Time-Dependent Ornstein-Uhlenbeck Process: a Model for Stochastic Decision-Making Process	May 2016

Scientific Community Services

Membership

• Member, ATLAS Collaboration , CERN	2024 – Present
• Affiliated member, Accelerated AI Algorithms for Data-Driven Discovery (A3D3) Institute	2024 – Present
• Member, CMS Collaboration , CERN	2020 – 2024

Collaboration Role

• Co-convener , SUSY Monte Carlo & Interpretation subgroup [CMS]	2023 – 2024
• Contact person , Higgs boson Monte Carlo group [CMS]	2021 – 2023
• Software developer , Data Quality Monitoring system for CaloLayer-1 trigger [CMS]	2020 – 2024

Journal Referee

Machine Learning and the Physical Sciences (2026); *Transactions on Machine Learning Research* (2026); *IEEE Transactions on Evolutionary Computation* (2026); *Wiley Advanced Theory and Simulations* (2025).

Teaching and Mentoring

Teaching

- Guest lecturer [UPenn], Data Analysis for the Natural Sciences II: Machine Learning (PHYS 3359, Spring 2025)
- Teaching assistant [U.Wisconsin], undergraduate physics courses (Fall 2018 – Spring 2020)

Master students

- Kam Wai Lai [UPenn]: Spiking neural network on FPGAs for cluster counting in drift chambers (2025–2026)

- Yuyan Wang [UPenn]: Joint-Embedding Predictive Architecture (JEPAs) for jet anomaly detection (2025–2026)

Undergraduate students

- Chenjia Ni [UPenn]: Lorentz-equivariant architecture on FPGAs (2026–Present)
- Abhay Agarwal [UPenn]: Scalable sparse convolution on FPGAs (2026–Present)
- Alex Yang [UPenn]: Mass regression of supersymmetric particles with contrastive learning (2025)
- Ashni Kumar [Drexel]: Acceleration of reinforcement learning agent inference on FPGAs for astronomical observations (2025)